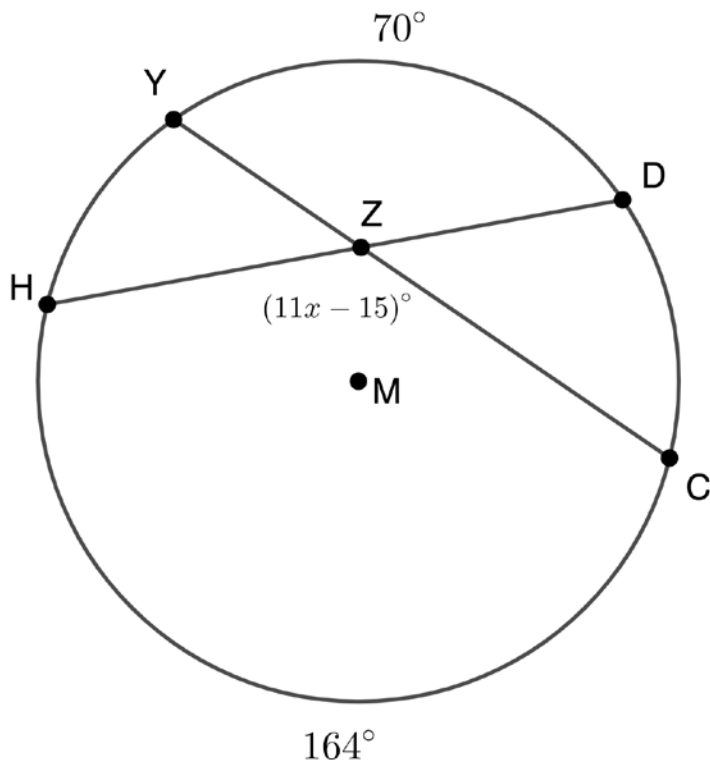


Geometry
Unit 10
Lesson 7 Practice

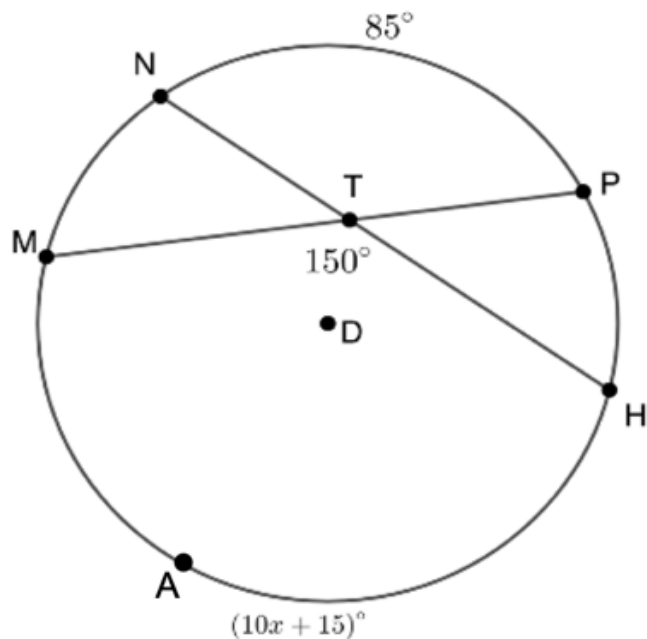
Name _____ **Answer Key** _____
Date _____
Period _____

For questions 1-3, Circle M is shown, where chords $\overline{YC}$ and $\overline{HD}$ intersect at point Z , $m\widehat{CH} = 164^\circ$, $m\widehat{YD} = 70^\circ$, and $m\angle HZC = (11x - 15)^\circ$.



- Find the $m\angle HZC$.
 $m\angle HZC = \frac{164+70}{2} = 117^\circ$
- Write an equation to find the value of x .
 $11x - 15 = 117$
- Find the value of x .
 $11x - 15 = 117$
 $11x = 132$
 $x = 12$

For questions 4-6, Circle D is shown, where chords $\overline{NH}$ and $\overline{MP}$ intersect at point T , $m\widehat{NP} = 85^\circ$, $m\widehat{MH} = (10x + 15)^\circ$, and $m\angle MTH = 150^\circ$.

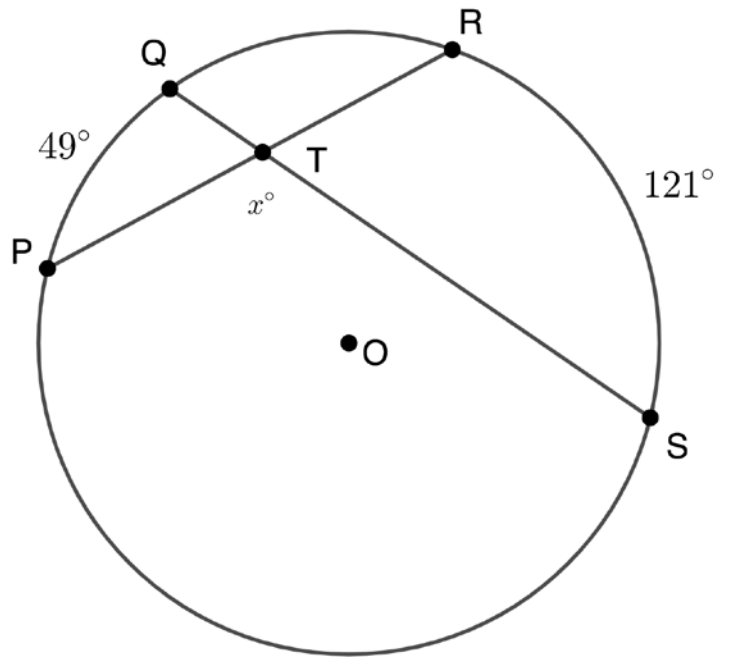


- Write an equation to find the value of x .
 $\frac{(10x+15)+85}{2} = 150$
- Find the value of x .
 $\frac{(10x+15)+85}{2} = 150$
 $10x + 100 = 300$
 $10x = 200$
 $x = 20$
- Find the $m\widehat{MH}$.
 $m\widehat{MH} = 10(20) + 15 = 215^\circ$

7. Circle O is shown, where chords $\overline{QS}$ and $\overline{RP}$ intersect at point T . Find the value of x for $\angle PTS$, given $m\widehat{RS} = 121^\circ$ and $m\widehat{PQ} = 49^\circ$.

$$m\angle RTS = \frac{121+49}{2} = 85^\circ$$

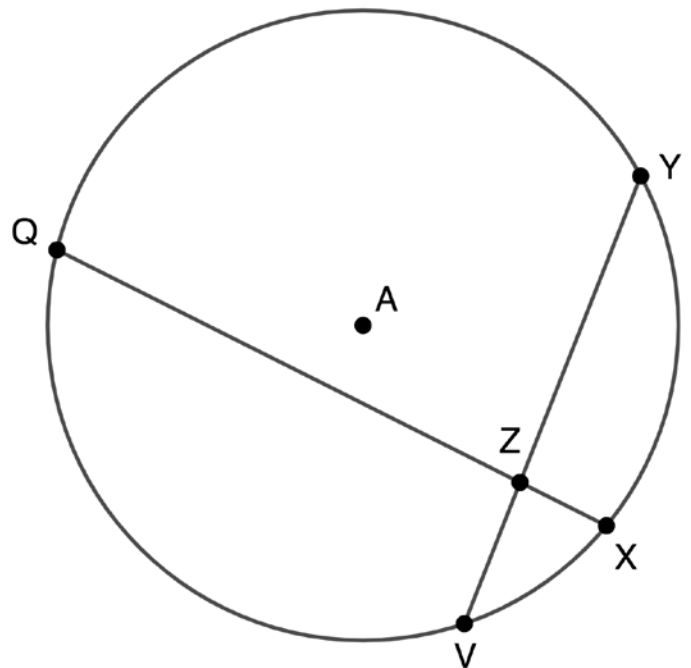
$$x = 180 - 85 = 95$$



For questions 8-10, Circle A is shown, where chords $\overline{QX}$ and $\overline{YV}$ intersect at point Z .

8. If $m\widehat{QY} = 137^\circ$ and $m\widehat{VX} = 35^\circ$, what is the measure of $\angle QZY$?

$$m\angle QZY = \frac{137+35}{2} = 86^\circ$$



9. If $m\widehat{QV} = 141^\circ$ and $m\widehat{YX} = 47^\circ$, what is the measure of $\angle QZV$?

$$m\angle QZV = \frac{141+47}{2} = 94^\circ$$

10. If $m\widehat{VX} = 41^\circ$ and $m\angle QZY = 84^\circ$, what is the measure of $\widehat{QY}$?

$$84 = \frac{41+m\widehat{QY}}{2}$$

$$168 = 41 + m\widehat{QY}$$

$$m\widehat{QY} = 127^\circ$$